Anant A. Joshi

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Summary

- As a control theorist and applied mathematician, my foundation is in **stochastic differential equations, stochastic optimal control (SOC), and filtering.** I am interested in problems in **sampling, SOC, generative models, and optimization**. **My research** focusses on **constructing sampling-based algorithms for SOC and reinforcement learning (RL) using interacting particle systems**.

Education

Ph.D. in Mech. Engg., Univ. of Illinois Urbana-Champaign (GPA 4.0/4.0, Advisor: Prof. Prashant Mehta) *Jan. 2021 - present*

- Awarded: Grainger Engineering Graduate Fellowship

B.Tech. in Mech. Engg. and M.Tech. in Systems and Control Engg., IIT Bombay, India (GPA 9.44/10) *Aug. 2015 - July 2020*

- Awarded: Best student paper award, 29th AAS/AIAA Space-Flight Mechanics Meeting, 2019

Patents

- S. Mowlavi, M. Benosman, **A. A. Joshi**, "*System and Method Suitable for Synthesizing a Control Policy for a Mechanical System*", USPTO App. 19/273,383

Experience

dual-EnKF Algorithm For Fast Simulation-Based Reinforcement Learning *UIUC*

Research with Prof. P. G. Mehta, Prof. A. Taghvaei, Prof. S. P. Meyn *Jan 2021 - Dec 2024*

- Developed a simulation based algorithm (dual-EnKF) for stochastic optimal control and RL in continuous-time continuous-state systems.
- Represented value function as a probability density and deployed the Ensemble Kalman Filter (EnKF) to sample efficiently from it.
- Numerically benchmarked dual-EnKF against policy gradient method, demonstrating upto 100x computational speedup.

Industry Internship: dual-EnKF for Data Driven Robust Control of PDEs *Mitsubishi Electric Research Labs*

Research intern with Dr. Mouhacine Benosman and Dr. Saviz Mowlavi *June 2024 - August 2024*

- Developed a data-driven, robust control algorithm for nonlinear systems governed by partial differential equations. Combined an optimal control estimated using dual-EnKF with a robust control obtained from Lyapunov redesign.
- Validated the controller's performance and robustness through extensive simulations on the heat equation and Burgers' equation.

Error Analysis of Sampling Algorithms for Stochastic Control *UIUC*

Research with Prof. P. G. Mehta and Prof. A. Taghvaei *Jan 2025 - ongoing*

- Developed an interacting particle-based sampling algorithm to numerically solve discrete-time stochastic optimal control. Analyzed the algorithm's approximation error and performance, and benchmarked it against traditional importance sampling methods.
- Highlighted scalability by demonstrating that the proposed algorithm avoids "curse of dimensionality" seen in importance sampling.

Observer Design for Systems on Smooth Manifolds with Lie Group Symmetry *IIT Bombay, India*

Research Project with Prof. R. N. Banavar and Prof. D. H. S. Maithripala *Aug. 2018 - Jul. 2019*

- Leveraged inherent geometric symmetry in nonlinear mechanical systems to simplify design observer into two smaller subsystems.

Uncertainty Quantification in Dynamical Systems with Invariants (AAS/AIAA Best paper) *University of Texas at Arlington*

Research Internship with Prof. K. S. Subbarao *May 2018 - July 2018*

- Modeled impact of Gaussian white noise on dynamical systems to quantify uncertainty in conserved quantities (invariants).
- Characterized the temporal evolution of the statistical moments of invariants. Theoretical results were verified with Monte Carlo simulation.

Leadership Roles

Head, Attitude Determination and Control Subsystem, Student Satellite Project *IIT Bombay*

- Responsible for envisioning and executing technical work in the 10 member subsystem. Regularly presented progress review to the faculty mentors.
- Laid out Quality-Assurance practices to ensure categorical documentation of project progress.

Selected Publications

- **A. A. Joshi**, S. Mowlavi, M. Benosman, "*A Dual Ensemble Kalman Filter Approach to Robust Control of Nonlinear Systems: An Application to Partial Differential Equations*", accepted to IEEE Conference on Decision and Control (CDC), 2025
- **A. A. Joshi**, H.-S. Chang, A. Taghvaei, P. G. Mehta, S. P. Meyn, "*Interacting Particle Systems for Fast Linear Quadratic RL*", Learning for Dynamics & Control Conference (L4DC), 2024
- P. Katdare, **A. A. Joshi**, K. Driggs-Campbell, "*Towards Provable Log Density Policy Gradient*", Transactions on Machine Learning Research (TMLR), 2024
- **A. A. Joshi**, D. Chatterjee, R. N. Banavar, "*Robust Discrete-Time Pontryagin Maximum Principle on Matrix Lie Groups*", IEEE Transactions on Automatic Control, 2021
- **A. A. Joshi**, K. Subbarao, "*Uncertainty Quantification and Analysis of Dynamical Systems with Invariants*", AAS-19-368, 29th AAS/AIAA Space-Flight Mechanics Meeting, January, 2019

Technical Skils

Programming: Python, PyTorch, MATLAB (Simulink, Control System Toolbox, Optimization Toolbox), C++

Control & Estimation: Model Predictive Control (MPC), Sensor Fusion, State Estimation, Kalman Filters (EnKF, EKF, UKF)

Coursework: Optimization, Reinforcement Learning, Machine Learning for Signal Processing, Deep Generative Models, State Estimation, Adaptive Control, Optimal Control, Stochastic Control, Probability and Random Processes, Stochastic Differential Equations